A 10,000-Neuron V1 Laminar Column in `jaxfne`: Spectrolaminar Proxy Readouts under Deep-Layer Drive

Generated with `jaxfne` (`import jaxfne as jtfne`)

June 15, 2026

Abstract

We build a single primary visual cortex (V1) laminar column of $N = 10{,}000$ reduced Izhikevich emitters in `jaxfne`, simulate 1000 ms at $\Delta t = 0.1$ ms, and read out a 128-contact laminar field with the package's *spectrolaminar suite*. Deep layers (L5/L6) are sparser ($\sim 3\times$ lower neuronal density), more excitatory, and receive a tunable constant (DC) drive; the dense recurrent connectivity propagates deep activity to superficial layers, where faster PV-mediated inhibition supports relatively higher γ power. All field outputs are **proxy readouts** (`field_solver_status=linear_solver`, `field_claim_level=proxy_readout`, `physical_amplitude_calibrated=Fa`
none are physical EEG/MEG/LFP/CSD measurements. A companion notebook reproduces every figure.

Contents

1 Model and methods

Operator chain. Emitter → Source → FieldProxy (linear_solver) → Probe → Objective. The emitter is a reduced Izhikevich population; per-neuron membrane voltage evolves under

$$\frac{dv}{dt} = 0.04v^2 + 5v + 140 - u + I, \qquad \frac{du}{dt} = a(bv - u), \qquad \text{if } v \geq 30 \text{ mV}: \; v \leftarrow c, \; u \leftarrow u + d.$$

The time loop is integrated with `jax.lax.scan`; recurrent input is the dense synaptic matrix W. The laminar field proxy is the linear projection

$$\text{lfp\_proxy}_{k,c} = \sum_n S_{k,n} \, K_{c,n}, \qquad \text{csd\_proxy}_{k,c} = D_{zz} \, \text{lfp\_proxy}_{k,c},$$

with a Gaussian depth kernel K and a finite-difference depth operator D_{zz} (`jtfne.project_laminar_sources`, $n_{\text{contacts}} = 128$).

Laminar structure. Five layers (L1, L2/3, L4, L5, L6). Deep layers are thinner/sparser and more excitatory; superficial layers carry more PV. Table 1 lists the realized per-layer counts and E-fractions; the constants appear in the companion notebook.

Deep drive. A constant current DC_{deep} is added to all L5/L6 cells (`model.with_emitter_parameters(drive_p`
Because W is dense, deep activity drives superficial populations through deep→superficial connections.

Table 1: Per-layer neuron counts and excitatory fraction (realized).

Layer	n	E-fraction
L1	1200	0.50
L2/3	3400	0.72
L4	2000	0.78
L5	1400	0.90
L6	2000	0.92

2 Cell densities per layer

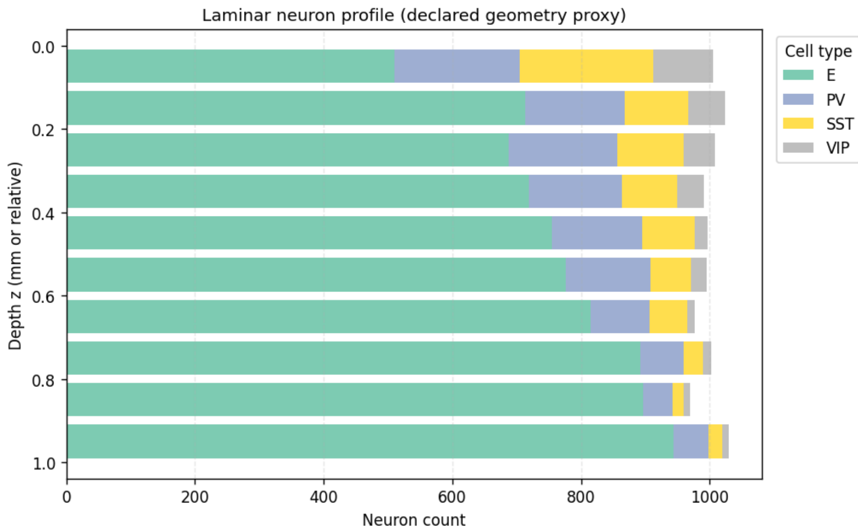

Figure 1: Per-layer cell-type composition. Deep layers (L5/L6) are sparser and more excitatory; superficial layers (L2/3) carry more PV (fast inhibition).

3 3D circuit of the 10,000-neuron V1 column

The interactive Plotly circuit is in `figs/09_circuit3d_10k.html` (pan/zoom/rotate). A static snapshot:

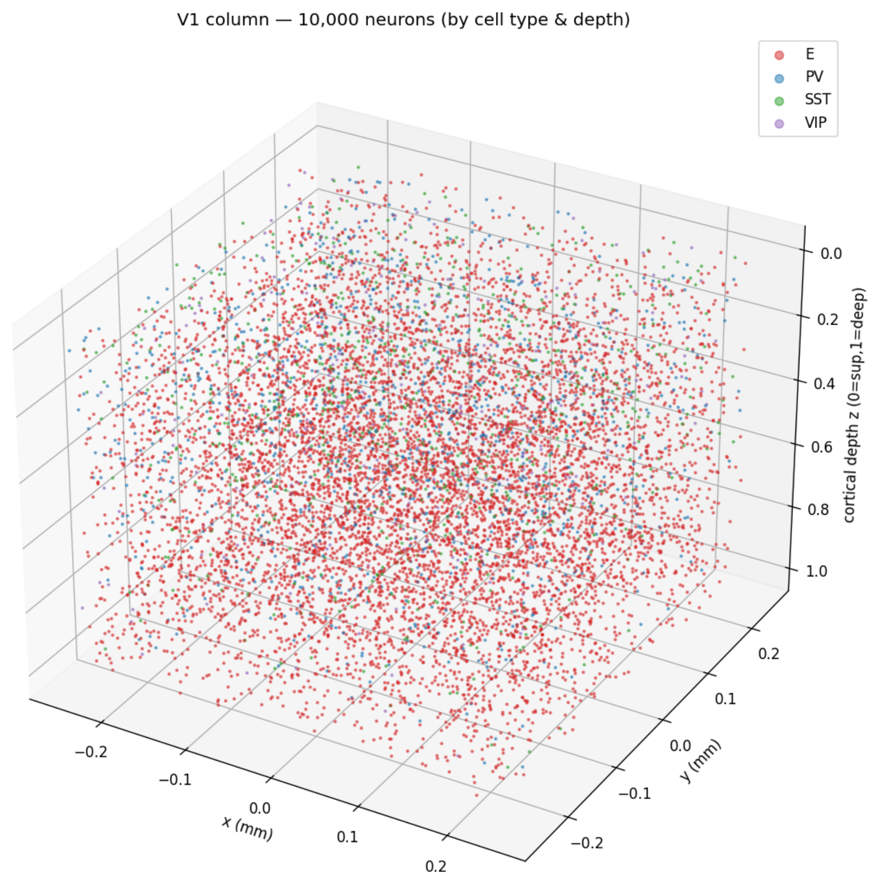

Figure 2: 3D layout of all 10,000 V1 neurons, colored by cell type, arranged by cortical depth (z). Superficial (top) to deep (bottom).

4 Firing rate and rasters

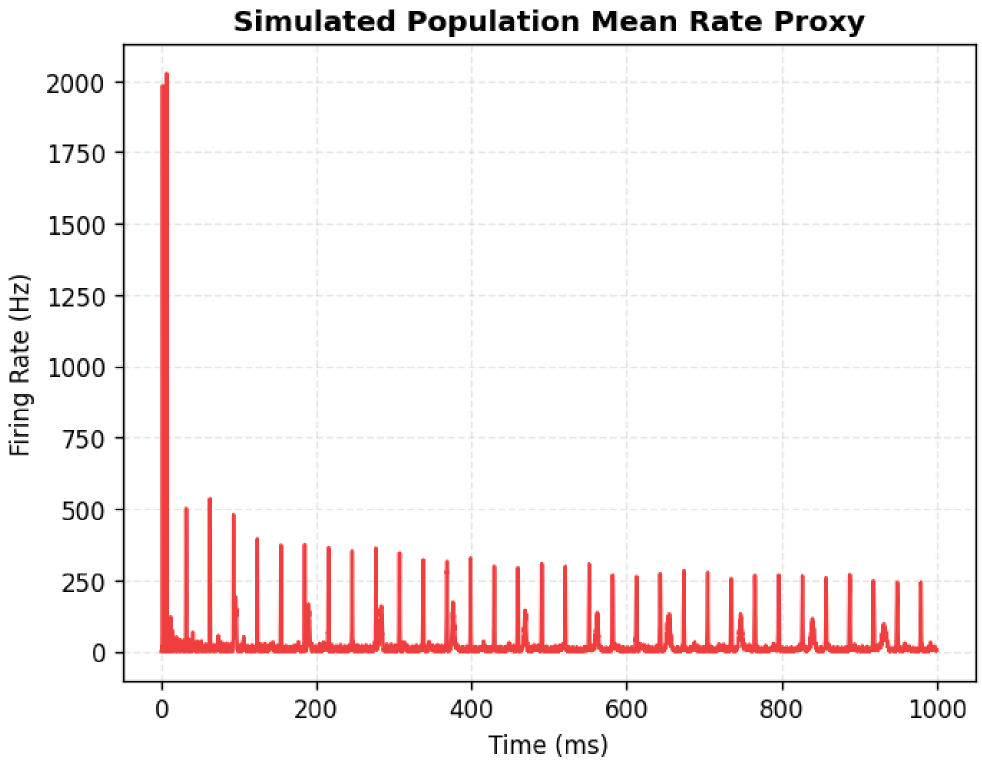

Figure 3: Population firing-rate proxy over the 1000 ms run at the main deep drive level.

4.1 Rasters across deep-drive levels

Increasing DC_{deep} raises deep firing and, via deep$\rightarrow$superficial connections, superficial activity.

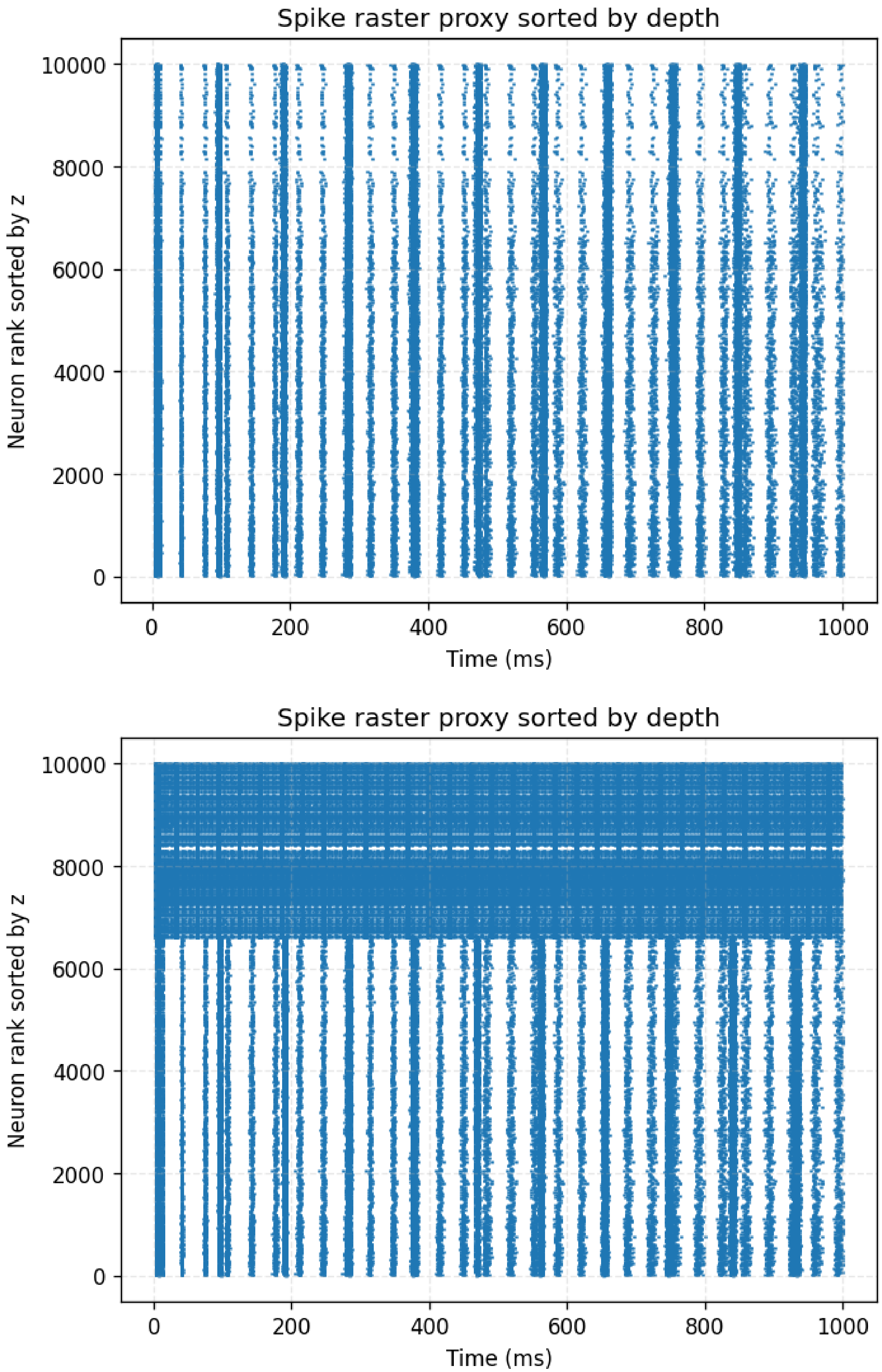

Figure 4: Spike rasters at $\mathrm{DC}_{\mathrm{deep}} = 0$ (top, mean rate 10.8 Hz) and $\mathrm{DC}_{\mathrm{deep}} = 10$ (bottom, 22.0 Hz). Deep drive raises deep firing and, through deep→superficial connections, superficial activity.

5 LFP proxy

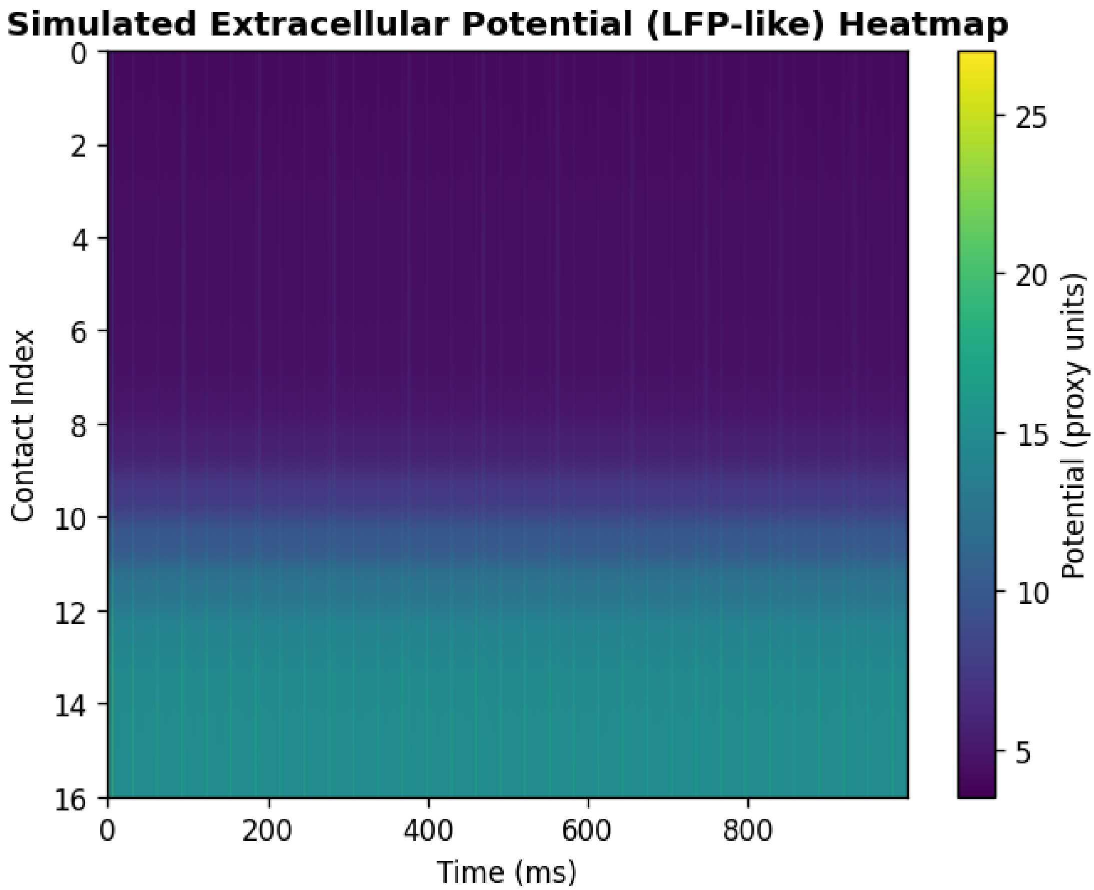

Figure 5: lfp_proxy across the 128 laminar contacts (relative units).

6 CSD proxy

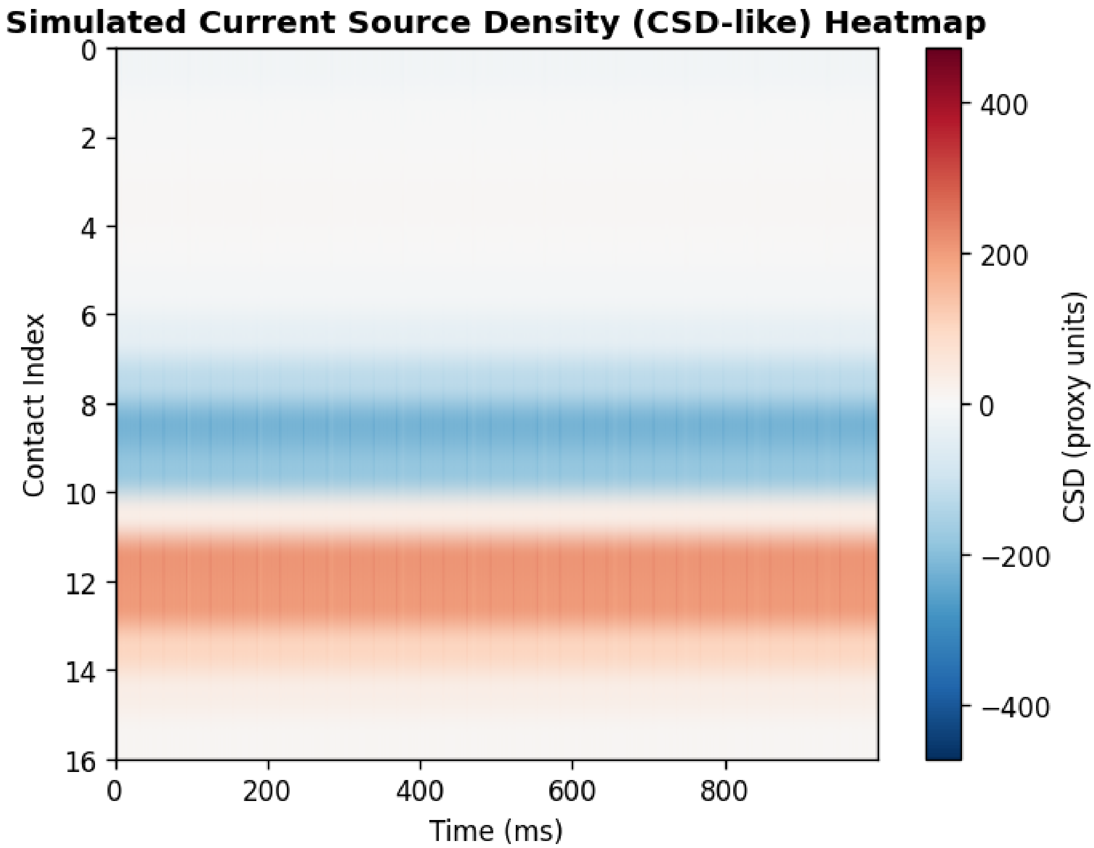

Figure 6: csd_proxy = D_{zz} lfp_proxy: current-source-density proxy (sink/source pattern), relative units.

7 Spectrolaminar suite (preferred readout)

The package spectrolaminar suite (`jtfne.vis.spectrolaminar_suite`) summarizes the depth×frequency relative-power structure and the α/β vs γ laminar band profiles.

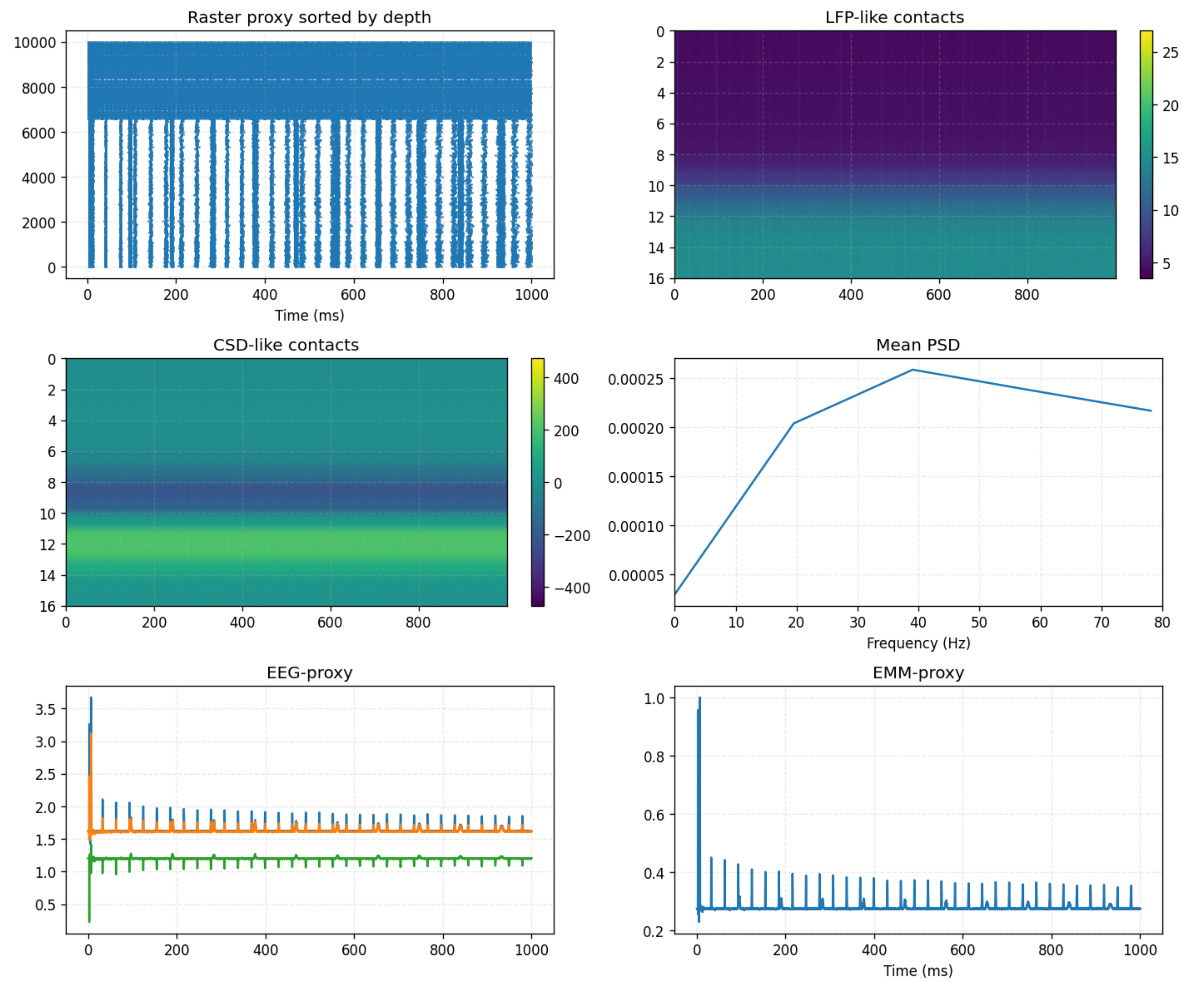

Figure 7: Spectrolaminar suite at the main deep-drive level: α/β (8–30 Hz) and γ (30–80 Hz) laminar profiles, depth$\times$frequency map.

7.1 Spectrolaminar suites across deep-drive levels

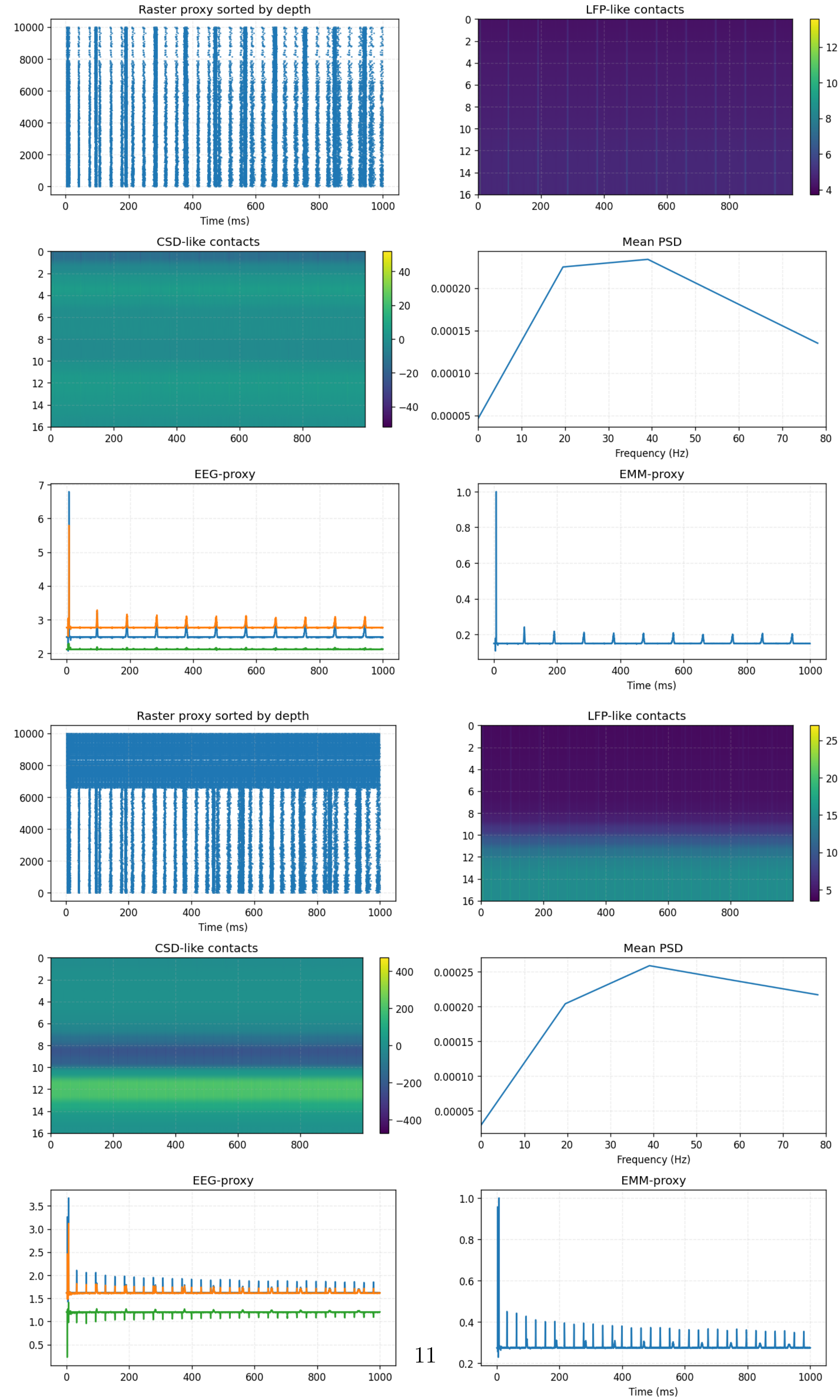

Figure 8: Spectrolaminar suite at deep drive $\mathrm{DC}_{\mathrm{deep}} = 0$ (top) and $\mathrm{DC}_{\mathrm{deep}} = 10$ (bottom). Increasing deep

8 EEG and MEG proxies

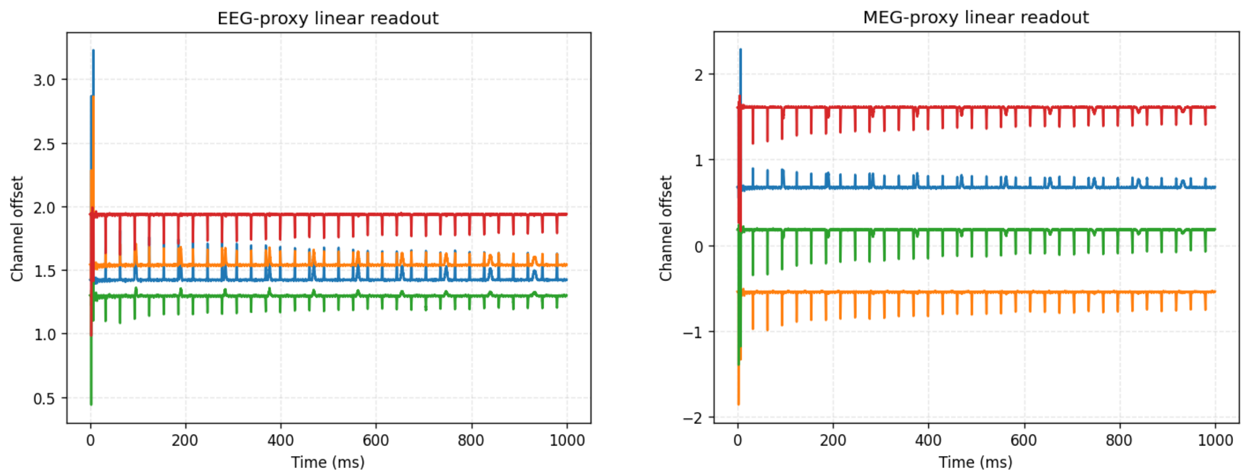

Figure 9: Left: eeg_proxy (linear leadfield projection to scalp electrodes). Right: meg_proxy (oriented-source magnetic projection). Both are proxy readouts in relative units.

9 Additional diagnostics

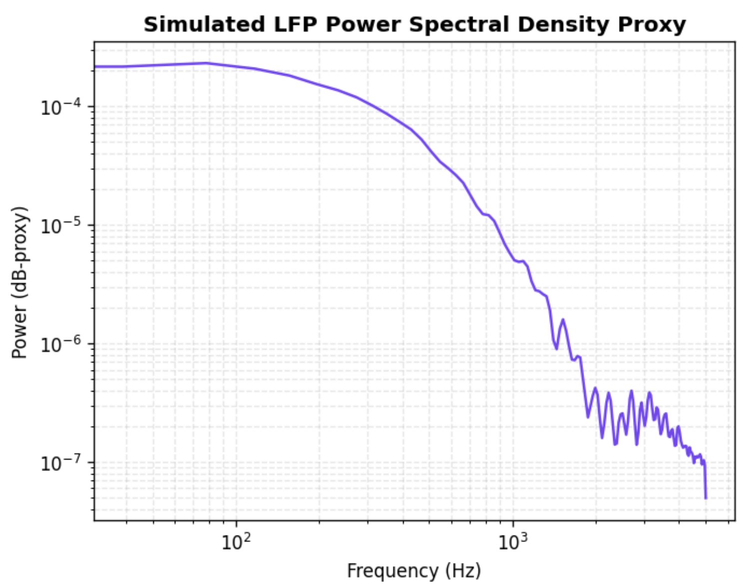

Figure 10: Power spectral density proxy of the laminar field.

10 Why the spectrolaminar suite is the preferred visualization

The spectrolaminar suite normalizes each band across depth, exposing the *relative* laminar organization (deep α/β, superficial γ) that raw LFP amplitude—dominated by the densest/most-driven layers—obscures. It is the standard readout for the laminar spectral motif and is robust to absolute-amplitude scaling, which is appropriate here because all outputs are uncalibrated proxies.

Scope statement

All field outputs are computational proxies (`claim_level=computational_scaffold`, `field_solver_status=linea`
`field_claim_level=proxy_readout`, `physical_amplitude_calibrated=False`). They are not physical EEG/MEG/LFP/CSD measurements and carry no calibrated amplitude.